Evaluating Algebraic Expressions

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Algebraic Expression

An **algebraic expression** is a mathematical phrase made up of numbers, variables, and operations.

$3x + 5$ $2a - 7$ $4m + 2n$

Variable

A **variable** is a letter or symbol that represents a number.

$$x \qquad y \qquad a$$

In $3x + 5$, the variable is x.

Parts of an Expression

$$3x + 5$$

- 3 number multiplying the variable
- x variable
- $+$ operation
- 5 number by itself

Evaluating an Expression

Evaluating an algebraic expression means finding its numerical value. Replace each variable with its given value, then perform the indicated operations.

Substitution

Substitution means replacing a variable with its given numerical value.

$$3x + 5 \quad \text{if} \quad x = 4$$

$$3(4) + 5$$

Negative Numbers

Use **parentheses** when substituting negative numbers to avoid errors.

$$x^2 \quad \text{if} \quad x = -3$$

$$(-3)^2 = 9$$

Without parentheses, -3^2 would give -9. That is wrong!

Steps in Evaluating

1. **Write** the expression.
2. **Substitute** the value of each variable. Use parentheses.
3. **Simplify** to get the final value.

Remember: **W-S-S**

Examples

Example 1

Evaluate $3x + 5$ if $x = 4$.

$$3(4) + 5$$

$$12 + 5$$

$$17$$

Example 2

Evaluate $4m + 2n$ if $m = 3$ and $n = 5$.

$$4(3) + 2(5)$$

$$12 + 10$$

$$22$$

Example 3

Evaluate $x^2 + 3x$ if $x = -2$.

$$(-2)^2 + 3(-2)$$

$$4 + (-6)$$

$$-2$$

Example 4

Evaluate $\dfrac{x+6}{3}$ if $x = 9$.

$$\frac{9+6}{3}$$

$$\frac{15}{3}$$

$$5$$

Practice Exercises

Practice 1

Evaluate $4x + 7$ if $x = 3$.

Practice 2

Evaluate $5a - 9$ if $a = 4$.

Practice 3

Evaluate $3y^2 + 2$ if $y = 2$.

Practice 4

Evaluate $6m - 2n$ if $m = 5$ and $n = 4$.

Practice 5

Evaluate $2p + 3q - 5$ if $p = 3$ and $q = 6$.

Practice 6

Evaluate $x^2 + 4x$ if $x = -3$.

Practice 7

Evaluate $3a^2 - 2b$ if $a = 2$ and $b = 5$.

Practice 8

Evaluate $\dfrac{x+8}{2}$ if $x = 6$.

Practice 9

Evaluate $2(3y - 4) + 5$ if $y = 3$.

Practice 10

Evaluate $\dfrac{2a+b}{3}$ if $a = 4$ and $b = 7$.

Activities

Activity 1

Evaluate $5x + 6$ if $x = 4$.

Activity 2

Evaluate $7a - 8$ if $a = 3$.

Activity 3

Evaluate $4y^2 + 1$ if $y = 3$.

Activity 4

Evaluate $5m - 3n$ if $m = 6$ and $n = 2$.

Activity 5

Evaluate $4p + 2q - 7$ if $p = 2$ and $q = 5$.

Activity 6

Evaluate $x^2 + 5x$ if $x = -4$.

Activity 7

Evaluate $2a^2 - 3b$ if $a = 3$ and $b = 4$.

Activity 8

Evaluate $\dfrac{x+10}{2}$ if $x = 8$.

Activity 9

Evaluate $3(2y - 5) + 4$ if $y = 4$.

Activity 10

Evaluate $\dfrac{3a + 2b}{4}$ if $a = 4$ and $b = 6$.

Thank you!